# JONATHAN JONGBIN KIM

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## EDUCATION

**Korea Advanced Institute of Science and Technology (KAIST)** Daejeon, Republic of Korea
*B.S. in Electrical Engineering* *Mar. 2025 – Present*

**GPA:** 3.65 / 4.3 | **Double Major:** Computer Science (in progress)
**Relevant Coursework (completed and in progress):** Introduction to Reinforcement Learning, Introduction to Generative Models, Machine Learning Basics and Practices, Probability and Introductory Random Processes, Programming Languages, Analysis I, Linear Algebra, Data Structures and Algorithms, Signals and Systems

## RESEARCH INTERESTS

Foundations of machine learning, learning and reasoning, world models, and reinforcement learning.

## MAJOR PROJECTS

**dongne-rec — Hyperlocal Sequential Recommendation** [GitHub] [Demo] 2026

- Built SASRec and four offline ranking protocols on 618K Foursquare check-ins from New York and Tokyo. In per-user leave-one-out, SASRec led with sampled candidates, while a visit-frequency baseline led on all candidates within 3 km.
- Built a FastAPI inference prototype. ONNX INT8 reduced the encoder from 13.4 to 3.5 MB. Local HTTP benchmarking measured p99 of 8.51 ms at concurrency 4 on an Apple M4 Pro.

**DiscoSpatial — Meta-Learned Updates for Spatial Memory** [GitHub] 2026

- Built a recurrent RGB Memory Maze benchmark comparing frozen Disco103 with V-trace actor-critic. Used 57 GPU pilot runs, synthetic positive controls, and reward-credit diagnostics to investigate near-uniform policies.

**FastGen — Few-Step Image Generation (Team Project)** 2026

- Proposed a GAN architecture and ideas for the training loss.
- Wrote the one-call checkpoint export and integrated the one- and four-call models under a shared evaluation interface. Co-wrote the team report.

**3D Gaussian Splatting — Scene Reconstruction** [GitHub] 2026

- Implemented a differentiable PyTorch renderer and a COLMAP-to-gsplat training pipeline with densification controls and TensorBoard logging.
- Trained the gsplat pipeline on 135 registered views for 30,000 steps on a Tesla T4, reaching 27.354 dB validation PSNR and 0.03535 L1 error.

## OTHER PROJECTS

**DiT-VITON — Virtual Try-On** [GitHub] 2026
Made latent resolution explicit in the DiT constructor and added patch divisibility and input shape checks. Exposed image size across preprocessing, training, and inference.

**BAGEL — Multimodal Inference and Image Editing** [GitHub] 2026
Built a CLI for ByteDance Seed's BAGEL-7B-MoT and measured foreground drift over 11 edits of a synthetic image.

**StarGAN v2 — Multi-Domain Image Translation** [GitHub] 2025
Trained a PyTorch implementation on AFHQ. FID 44.11 and LPIPS diversity 0.4270 on 3,000 samples.

## PUBLICATIONS & AWARDS

[1] **J. Kim**, et al., "*Analyzing The Impact and Evaluation of Flextime on Subway Congestion Using SARIMA Model*", *Proceedings of the Korea Computer Congress (KCC)*, 2024.

[2] **J. Kim**, et al., "*Comparison of Deep Learning Architecture Performance and Suggestion of Baseline Model for Diagnosing Cassava Leaf Disease*", *Proceedings of the Korea Software Congress (KSC)*, 2024.

**Award** Excellence Award, Junior Division, KSC Paper Competition (2024).

## SKILLS

**Programming:** Python, C++, C, Scala, Assembly, Bash | **ML:** PyTorch, NumPy, ONNX Runtime
**Tools:** Git, Linux, CUDA, LaTeX